

Christian Lentz

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Revised: June 2026

Education

Tufts University

MS, Mathematics

Medford, MA

08/2025 - present

- Advisor: [Abiy Tasissa](#)

Macalester College

BA, Mathematics and Computer Science

St. Paul, MN

05/2024

- Summa Cum Laude
- Advisors: [Lori Ziegelmeier](#) and [Susan Fox](#)
- Honors Thesis: [Persistent Relative Homology for Topological Data Analysis](#)

Oxford University

Visiting Student, Mathematics

Oxford, England, UK

01/2023 - 06/2023

Interests

Mathematics – Topology, Geometry, Homological algebra

Applied Mathematics – Numerical analysis & linear algebra, Computational geometry, Optimization

Algorithms and Machine Learning – Topological data analysis, Compressing sensing, Dictionary learning, Algorithm unrolling, Complexity theory

Research

University of Minnesota / Macalester College

Research Assistant

St. Paul, MN

05/2023 - 08/2025

- Algebraic and computational topology, applied homological algebra, lazy/sparse matrix algebra.
- Developed a novel algorithm to compute persistent relative homology which provides cycle representatives, persistence modules and barcode decompositions.
- Advisors: Lori Ziegelmeier (Macalester Col.) and Gregory Henselman-Petrusek (PNNL).

Teaching

Tufts University

Gader

Medford, MA

08/2025 - 12/2025

- Responsibilities: Grade homework, proctor exams.
- *Mathematical Aspects of Data Analysis*, Fall 2025

Macalester College

Teaching Assistant

St. Paul, MN

01/2022 - 05/2024

- Responsibilities: Attend lectures, design and grade homework, hold office hours twice weekly.
- *Linear Algebra*, Spring 2022
- *Introduction to Statistical Modeling*, Fall 2022
- *Computational Geometry*, Fall 2023
- *Algorithm Design and Analysis*, Spring 2024

Publications

Lentz, C., Henselman-Petrusek G., Ziegelmeier L. (preprint). A U-match Algorithm for Persistent Relative Homology. [arXiv:2602.03163](https://arxiv.org/abs/2602.03163).

Lentz, C. (2024). Persistent Relative Homology for Topological Data Analysis. *Mathematics, Statistics, and Computer Science Honors Projects*. 85. https://digitalcommons.macalester.edu/mathcs_honors/85.

Talks & Presentations

Invited Talks

2024 January: Joint Mathematics Meetings, AIM-AMS Special Session on Applied Topology Beyond Persistence Diagrams, *A computational approach for persistent relative homology*.

Contributed Talks

2023 September: Fall Meeting of Mathematical Association of America NCS, *A matrix factorization algorithm for persistent relative homology*.

Undergraduate Sessions

2023 October: Macalester College, Summer Showcase Seminar, *A matrix factorization algorithm for persistent relative homology*.

2024 January: Joint Mathematics Meetings, PME Undergraduate Student Poster Session, *A computational approach for persistent relative homology*.

2024 April: Undergraduate honors defense, Macalester College, Department of Mathematics, Statistics and Computer Science, *Persistent relative homology for topological data analysis*.

Awards

Konhauser Achievement Award

Macalester College, 2024

- Awarded each year to a single student majoring in mathematics at Macalester College for outstanding academic record and demonstrated dedication to and interest in the field.

Dewitt Wallace Distinguished Scholarship

Macalester College, 2020 - 2024

- Based on academic merit and awarded on a highly-competitive basis.

Open Source Code

Open Applied Topology (in progress)

[Repository](#)

Contributions:

- A Python implementation for computing persistent relative homology from point cloud data which utilizes lazy matrix structures and the [U-match](#) factorization scheme.
- Visualization tools for cycle representatives and barcodes.

Other Experience

Maverick Software Consulting

Minneapolis, MN

QA Software Engineer, Internship

06/2022 - 01/2023

- Supervisor: [Tracy Olhausen](#), Senior Director of Quality Assurance.

Relevant Skills

Languages | Python, Java, JavaScript, R, Rust, C

Software Engineering | HTML, CSS, Node.js, Google Firebase

ML & Data Science | NumPy, SciPy, PyTorch, Plotly, matplotlib, RStudio, Tidyverse, numerical algorithms

Misc. Technologies | VS Code, Git/GitHub, Mathematica, Jira

Research & Teaching | literature review, technical & academic writing, project management, collaboration